What's new in Kotlin 2.1.20

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The Kotlin 2.1.20 release is here! Here are the main highlights:

- **K2 compiler updates**: Updates to the new kapt and Lombok plugins
- **Kotlin Multiplatform**: New DSL to replace Gradle's Application plugin
- **Kotlin/Native**: Support for Xcode 16.3 and a new inlining optimization
- **Kotlin/Wasm**: Default custom formatters, support for DWARF, and migration to Provider API
- **Gradle support**: Compatibility with Gradle's Isolated Projects and custom publication variants
- **Standard library**: Common atomic types, improved UUID support, and new time-tracking functionality
- **Compose compiler**: Relaxed restrictions on `@Composable` functions and other updates
- **Documentation**: Important improvements to the Kotlin documentation

For information about the Kotlin release cycle, see Kotlin release process.

IDE support

The Kotlin plugins that support 2.1.20 are bundled in the latest IntelliJ IDEA and Android Studio. You don't need to update the Kotlin plugin in your IDE. All you need to do is to change the Kotlin version to 2.1.20 in your build scripts.

See Update to a new release for details.

Download sources for Kotlin artifacts in projects with OSGi support

Sources of all dependencies of the `kotlin-osgi-bundle` library are now included in its distribution. This allows IntelliJ IDEA to download these sources to provide documentation for Kotlin symbols and improve the debugging experience.

Kotlin K2 compiler

We're continuing to improve plugin support for the new Kotlin K2 compiler. This release brings updates to the new kapt and Lombok plugins.

Beta

New default kapt plugin

Starting with Kotlin 2.1.20, the K2 implementation of the kapt compiler plugin is enabled by default for all the projects.

The JetBrains team launched the new implementation of the kapt plugin with the K2 compiler back in Kotlin 1.9.20. Since then, we have further developed the internal implementation of K2 kapt and made its behavior similar to that of the K1 version, while significantly improving its performance as well.

If you encounter any issues when using kapt with the K2 compiler, you can temporarily revert to the previous plugin implementation.

To do this, add the following option to the `gradle.properties` file of your project:

```
kapt.use.k2=false
```

Please report any issues to our issue tracker ↗.

Experimental

Lombok compiler plugin: support for `@SuperBuilder` and updates on `@Builder`

The Kotlin Lombok compiler plugin now supports the `@SuperBuilder` annotation, making it easier to create builders for class hierarchies. Previously, developers using Lombok in Kotlin had to manually define builders when working with inheritance. With `@SuperBuilder`, the builder automatically inherits superclass fields, allowing you to initialize them when constructing an object.

Additionally, this update includes several improvements and bug fixes:

- The `@Builder` annotation now works on constructors, allowing more flexible object creation. For more details, see the corresponding YouTrack issue ↗.

- Several issues related to Lombok's code generation in Kotlin have been resolved, improving overall compatibility. For more details, see the GitHub changelog ↗.

For more information about the `@SuperBuilder` annotation, see the official Lombok documentation ↗.

Experimental

Kotlin Multiplatform: new DSL to replace Gradle's Application plugin

Starting with Gradle 8.7, the Application ↗ plugin is no longer compatible with the Kotlin Multiplatform Gradle plugin. Kotlin 2.1.20 introduces an Experimental DSL to achieve similar functionality. The new `executable {}` block configures execution tasks and Gradle distributions ↗ for JVM targets.

Before the `executable {}` block in your build script, add the following `@OptIn` annotation:

```
@OptIn(ExperimentalKotlinGradlePluginApi::class)
```

For example:

```
kotlin {
    jvm {
        @OptIn(ExperimentalKotlinGradlePluginApi::class)
        binaries {
            // Configures a JavaExec task named "runJvm" and a Gradle dist
            executable {
                mainClass.set("foo.MainKt")
            }

            // Configures a JavaExec task named "runJvmAnother" and a Grac
            executable(KotlinCompilation.MAIN_COMPILATION_NAME, "another")
                // Set a different class
                mainClass.set("foo.MainAnotherKt")
            }

            // Configures a JavaExec task named "runJvmTest" and a Gradle
            executable(KotlinCompilation.TEST_COMPILATION_NAME) {
                mainClass.set("foo.MainTestKt")
            }

            // Configures a JavaExec task named "runJvmTestAnother" and a
            executable(KotlinCompilation.TEST_COMPILATION_NAME, "another")
                mainClass.set("foo.MainAnotherTestKt")
            }
        }
```

```
    }
}
```

In this example, Gradle's Distribution ↗ plugin is applied on the first `executable {}` block.

If you run into any issues, report them in our issue tracker ↗ or let us know in our public Slack channel ↗.

Kotlin/Native

Support for Xcode 16.3

Starting with Kotlin **2.1.21**, the Kotlin/Native compiler supports Xcode 16.3 – the latest stable version of Xcode. Feel free to update your Xcode and continue working on your Kotlin projects for Apple operating systems.

The 2.1.21 release also fixes the related cinterop issue ↗ that caused compilation failures in Kotlin Multiplatform projects.

Experimental

New inlining optimization

Kotlin 2.1.20 introduces a new inlining optimization pass, which comes before the actual code generation phase.

The new inlining pass in the Kotlin/Native compiler should perform better than the standard LLVM inliner and improve the runtime performance of the generated code.

The new inlining pass is currently Experimental. To try it out, use the following compiler option:

```
-Xbinary=preCodegenInlineThreshold=40
```

Our experiments show that setting the threshold to 40 tokens (code units parsed by the compiler) provides a reasonable compromise for compilation optimization. According to our benchmarks, this gives an overall performance improvement of 9.5%. Of course, you can try out other values, too.

If you experience increased binary size or compilation time, please report such issues via YouTrack ↗.

Kotlin/Wasm

This release improves Kotlin/Wasm debugging and property usage. Custom formatters now work out of the box in development builds, while DWARF debugging facilitates code inspection. Additionally, the Provider API simplifies property usage in Kotlin/Wasm and Kotlin/JS.

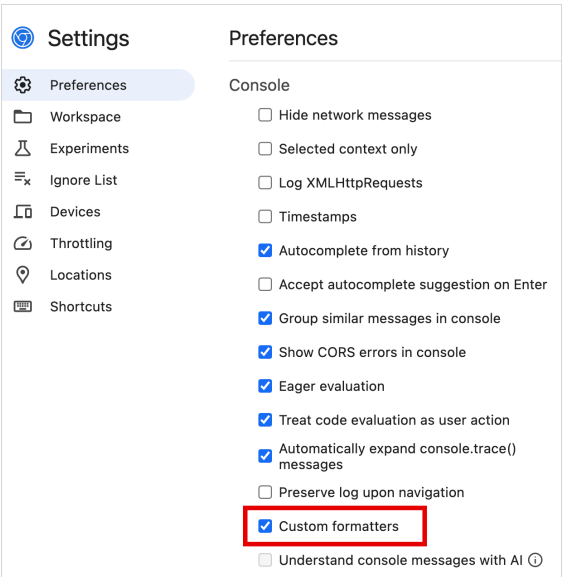
Custom formatters enabled by default

Before, you had to manually configure custom formatters to improve debugging in web browsers when working with Kotlin/Wasm code.

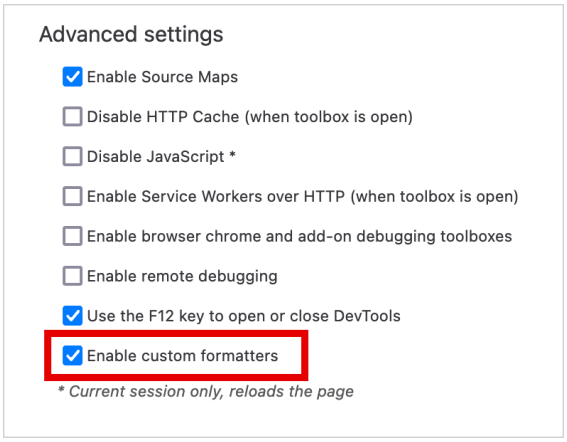
In this release, custom formatters are enabled by default in development builds, so you don't need additional Gradle configurations.

To use this feature, you only need to ensure that custom formatters are enabled in your browser's developer tools:

- In Chrome DevTools, find the custom formatters checkbox in **Settings | Preferences | Console**:



- In Firefox DevTools, find the custom formatters checkbox in **Settings | Advanced settings**:



This change primarily affects Kotlin/Wasm development builds. If you have specific requirements for production builds, you need to adjust your Gradle configuration accordingly. To do so, add the following compiler option to the `wasmJs {}` block:

```
// build.gradle.kts
kotlin {
    wasmJs {
        // ...

        compilerOptions {
            freeCompilerArgs.add("-Xwasm-debugger-custom-formatters")
        }
    }
}
```

Support for DWARF to debug Kotlin/Wasm code

Kotlin 2.1.20 introduces support for DWARF (debugging with arbitrary record format) in Kotlin/Wasm.

With this change, the Kotlin/Wasm compiler is able to embed DWARF data into the generated WebAssembly (Wasm) binary. Many debuggers and virtual machines can read this data to provide insights into the compiled code.

DWARF is mainly useful for debugging Kotlin/Wasm applications inside standalone Wasm virtual machines (VMs). To use this feature, the Wasm VM and debugger must support DWARF.

With DWARF support, you can step through Kotlin/Wasm applications, inspect variables, and gain code insights. To enable this feature, use the following compiler option:

```
-Xwasm-generate-dwarf
```

Migration to Provider API for Kotlin/Wasm and Kotlin/JS properties

Previously, properties in Kotlin/Wasm and Kotlin/JS extensions were mutable (`var`) and assigned directly in build scripts:

```
the<NodeJsExtension>().version = "2.0.0"
```

Now, properties are exposed through the Provider API ↗, and you must use the `.set()` function to assign values:

```
the<NodeJsEnvSpec>().version.set("2.0.0")
```

The Provider API ensures that values are lazily computed and properly integrated with task dependencies, improving build performance.

With this change, direct property assignments are deprecated in favor of `*EnvSpec` classes, such as `NodeJsEnvSpec` and `YarnRootEnvSpec`.

Additionally, several alias tasks have been removed to avoid confusion:

Deprecated task	Replacement
`wasmJsRun`	`wasmJsBrowserDevelopmentRun`
`wasmJsBrowserRun`	`wasmJsBrowserDevelopmentRun`
`wasmJsNodeRun`	`wasmJsNodeDevelopmentRun`
`wasmJsBrowserWebpack`	`wasmJsBrowserProductionWebpack` or `wasmJsBrowserDistribution`
`jsRun`	`jsBrowserDevelopmentRun`
`jsBrowserRun`	`jsBrowserDevelopmentRun`
`jsNodeRun`	`jsNodeDevelopmentRun`
`jsBrowserWebpack`	`jsBrowserProductionWebpack` or `jsBrowserDistribution`

If you only use Kotlin/JS or Kotlin/Wasm in build scripts, no action is required as Gradle automatically handles assignments.

However, if you maintain a plugin based on the Kotlin Gradle Plugin, and your plugin does not apply `kotlin-dsl`, you must update property assignments to use the `.set()` function.

Gradle

Kotlin 2.1.20 is fully compatible with Gradle 7.6.3 through 8.11. You can also use Gradle versions up to the latest Gradle release. However, be aware that doing so may result in deprecation warnings, and some new Gradle features might not work.

This version of Kotlin includes compatibility of Kotlin Gradle plugins with Gradle's Isolated Projects as well as support for custom Gradle publication variants.

Experimental

Kotlin Gradle plugins compatible with Gradle's Isolated Projects

> This feature is currently in a pre-Alpha state in Gradle. JS and Wasm targets are not supported at the moment. Use it only with Gradle version 8.10 or higher and solely for evaluation purposes.

Since Kotlin 2.1.0, you've been able to preview Gradle's Isolated Projects feature in your projects.

Previously, you had to configure the Kotlin Gradle plugin to make your project compatible with the Isolated Projects feature before you could try it out. In Kotlin 2.1.20, this additional step is no longer necessary.

Now, to enable the Isolated Projects feature, you only need to set the system property ↗.

Gradle's Isolated Projects feature is supported in Kotlin Gradle plugins for both multiplatform projects and projects that contain only the JVM or Android target.

Specifically for multiplatform projects, if you notice problems with your Gradle build after upgrading, you can opt out of the new Kotlin Gradle plugin behavior by adding:

```
kotlin.kmp.isolated-projects.support=disable
```

However, if you use this Gradle property in your multiplatform project, you can't use the Isolated Projects feature.

Let us know about your experience with this feature in YouTrack ↗.

Experimental

Support for adding custom Gradle publication variants

Kotlin 2.1.20 introduces support for adding custom Gradle publication variants ↗. This feature is available for multiplatform projects and projects targeting the JVM.

You cannot modify existing Gradle variants with this feature.

This feature is Experimental. To opt in, use the `@OptIn(ExperimentalKotlinGradlePluginApi::class)` annotation.

To add a custom Gradle publication variant, invoke the `adhocSoftwareComponent()` function, which returns an instance of `AdhocComponentWithVariants` ↗ that you can configure in the Kotlin DSL:

```
plugins {
    // Only JVM and Multiplatform are supported
    kotlin("jvm")
    // or
    kotlin("multiplatform")
}


kotlin {
    @OptIn(ExperimentalKotlinGradlePluginApi::class)
    publishing {
        // Returns an instance of AdhocSoftwareComponent
        adhocSoftwareComponent()
        // Alternatively, you can configure AdhocSoftwareComponent in the
        adhocSoftwareComponent {
            // Add your custom variants here using the AdhocSoftwareCompor
        }
    }
}
```

For more information on variants, see Gradle's Customizing publishing guide ↗.

Standard library

This release brings new Experimental features to the standard library: common atomic types, improved support for UUIDs, and new time-tracking functionality.

Experimental

Common atomic types

In Kotlin 2.1.20, we are introducing common atomic types in the standard library's `kotlin.concurrent.atomics` package, enabling shared, platform-independent code for thread-safe operations. This simplifies development for Kotlin Multiplatform projects by removing the need to duplicate atomic-dependent logic across source sets.

The `kotlin.concurrent.atomics` package and its properties are Experimental. To opt in, use the `@OptIn(ExperimentalAtomicApi::class)` annotation or the compiler option `-opt-in=kotlin.ExperimentalAtomicApi`.

Here's an example that shows how you can use `AtomicInt` to safely count processed items across multiple threads:

```
@OptIn(ExperimentalAtomicApi::class)
suspend fun main() {
    // Initializes the atomic counter for processed items
    var processedItems = AtomicInt(0)
    val totalItems = 100
    val items = List(totalItems) { "item$it" }
    // Splits the items into chunks for processing by multiple coroutines
    val chunkSize = 20
    val itemChunks = items.chunked(chunkSize)
    coroutineScope {
        for (chunk in itemChunks) {
            launch {
                for (item in chunk) {
                    println("Processing $item in thread ${Thread.currentT
                    processedItems += 1 // Increment counter atomically
                }
            }
        }
    }
```

Open in Playground → Target: JVM Running on v.2.3.20

To enable seamless interoperability between Kotlin's atomic types and Java's `java.util.concurrent.atomic` ↗ atomic types, the API provides the `.asJavaAtomic()` and `.asKotlinAtomic()` extension functions. On the JVM, Kotlin atomics and Java atomics are the same types in runtime, so you can transform Java atomics into Kotlin atomics and vice versa without any overhead.

Here's an example that shows how Kotlin and Java atomic types can work together:

```
@OptIn(ExperimentalAtomicApi::class)
fun main() {
    // Converts Kotlin's AtomicInt to Java's AtomicInteger
    val kotlinAtomic = AtomicInt(42)
    val javaAtomic: AtomicInteger = kotlinAtomic.asJavaAtomic()
    println("Java atomic value: ${javaAtomic.get()}")
    // Java atomic value: 42

    // Converts Java's AtomicInteger back to Kotlin's AtomicInt
    val kotlinAgain: AtomicInt = javaAtomic.asKotlinAtomic()
    println("Kotlin atomic value: ${kotlinAgain.load()}")
    // Kotlin atomic value: 42
}
```

Open in Playground → Target: JVM Running on v.2.3.20

Experimental

Changes in UUID parsing, formatting, and comparability

The JetBrains team continues to improve the support for UUIDs introduced to the standard library in 2.0.20.

Previously, the `parse()` function only accepted UUIDs in the hex-and-dash format. With Kotlin 2.1.20, you can use `parse()` for **both** the hex-and-dash and the plain hexadecimal (without dashes) formats.

We've also introduced functions specific to operations with the hex-and-dash format in this release:

- `parseHexDash()` parses UUIDs from the hex-and-dash format.
- `toHexDashString()` converts a `Uuid` into a `String` in the hex-and-dash format (mirroring the functionality of `toString()`).

These functions work similarly to `parseHex()` ↗ and `toHexString()` ↗, which were introduced earlier for the hexadecimal format. Explicit naming for parsing and formatting functionality should improve code clarity and your overall experience with UUIDs.

UUIDs in Kotlin are now `Comparable`. Starting with Kotlin 2.1.20, you can directly compare and sort values of the `Uuid` type. This enables the use of the `<` and `>` operators and standard library extensions available exclusively for `Comparable` types or their collections (such as `sorted()`), and it also allows passing UUIDs to any functions or APIs that require the `Comparable` interface.

Remember that the UUID support in the standard library is still Experimental. To opt in, use the `@OptIn(ExperimentalUuidApi::class)` annotation or the compiler option `-opt-in=kotlin.uuid.ExperimentalUuidApi`:

```
@OptIn(ExperimentalUuidApi::class)
fun main() {
    // parse() accepts a UUID in a plain hexadecimal format
    val uuid = Uuid.parse("550e8400e29b41d4a716446655440000")

    // Converts it to the hex-and-dash format
    val hexDashFormat = uuid.toHexDashString()

    // Outputs the UUID in the hex-and-dash format
    println(hexDashFormat)

    // Outputs UUIDs in ascending order
    println(
        listOf(
            uuid,
```

```
            Uuid.parse("780e8400e29b41d4a716446655440005"),
            Uuid.parse("5ab88400e29b41d4a716446655440076")
        ).sorted()
    )
}
```

Open in Playground → Target: JVM Running on v.2.3.20

Experimental

New time tracking functionality

Starting with Kotlin 2.1.20, the standard library provides the ability to represent a moment in time. This functionality was previously only available in `kotlinx-datetime` ↗, an official Kotlin library.

The `kotlinx.datetime.Clock` interface is introduced to the standard library as `kotlin.time.Clock` ↗ and the `kotlinx.datetime.Instant` class as `kotlin.time.Instant` ↗. These concepts naturally align with the `time` package in the standard library because they're only concerned with moments in time compared to a more complex calendar and timezone functionality that remains in `kotlinx-datetime`.

`Instant` and `Clock` are useful when you need precise time tracking without considering time zones or dates. For example, you can use them to log events with timestamps, measure durations between two points in time, and obtain the current moment for system processes.

To provide interoperability with other languages, additional converter functions are available:

- `.toKotlinInstant()` ↗ converts a time value to a `kotlin.time.Instant` instance.
- `.toJavaInstant()` ↗ converts the `kotlin.time.Instant` value to a `java.time.Instant` value.
- `Instant.toJSDate()` ↗ converts the `kotlin.time.Instant` value to an instance of the JS `Date` class. This conversion is not precise; JS uses millisecond precision to represent dates, while Kotlin allows for nanosecond resolution.

The new time features of the standard library are still Experimental. To opt in, use the `@OptIn(ExperimentalTime::class)` annotation:

```
import kotlin.time.*

@OptIn(ExperimentalTime::class)
fun main() {

    // Get the current moment in time
    val currentInstant = Clock.System.now()
    println("Current time: $currentInstant")

    // Find the difference between two moments in time
    val pastInstant = Instant.parse("2023-01-01T00:00:00Z")
    val duration = currentInstant - pastInstant
```

```
    println("Time elapsed since 2023-01-01: $duration")
}
```

Open in Playground → Target: JVM Running on v.2.3.20

For more information on the implementation, see this KEEP proposal ↗.

Compose compiler

In 2.1.20, the Compose compiler relaxes some restrictions on `@Composable` functions introduced in previous releases. In addition, the Compose compiler Gradle plugin is set to include source information by default, aligning the behavior on all platforms with Android.

Support for parameters with default values in open `@Composable` functions

The compiler previously restricted parameters with default values in open `@Composable` functions due to incorrect compiler output, which would result in crashes at runtime. The underlying issue is now resolved, and parameters with default values are fully supported when used with Kotlin 2.1.20 or newer.

Compose compiler allowed parameters with default values in open functions before version 1.5.8 ↗, so the support depends on project configuration:

- If an open composable function is compiled with Kotlin version 2.1.20 or newer, the compiler generates correct wrappers for parameters with default values. This includes wrappers compatible with pre-1.5.8 binaries, meaning that downstream libraries will also be able to use this open function.

- If the open composable function is compiled with Kotlin older than 2.1.20, Compose uses a compatibility mode, which might result in runtime crashes. When using the compatibility mode, the compiler emits a warning to highlight potential problems.

Final overridden functions are allowed to be restartable

Virtual functions (overrides of `open` and `abstract`, including interfaces) were forced to be non-restartable with the 2.1.0 release. This restriction is now relaxed for functions that are members of final classes or are `final` themselves – they will be restarted or skipped as usual.

You might observe some behavior changes in affected functions after upgrading to Kotlin 2.1.20. To force non-restartable logic from the previous version, apply the `@NonRestartableComposable` annotation to the function.

`ComposableSingletons` removed from public API

`ComposableSingletons` is a class created by the Compose compiler when optimizing `@Composable` lambdas. Lambdas that do not capture any parameters are allocated once and cached in a property of the class, saving allocations during runtime. The class is generated with internal visibility and is only intended for optimizing lambdas inside a compilation unit (usually a file).

However, this optimization was also applied to `inline` function bodies, which resulted in singleton lambda instances leaking into the public API. To fix this problem, starting with 2.1.20, `@Composable` lambdas are no longer optimized into singletons inside inline functions. At the same time, the Compose compiler will continue generating singleton classes and lambdas for inline functions to support binary compatibility for modules that were compiled under the previous model.

Source information included by default

The Compose compiler Gradle plugin already has the including source information ↗ feature enabled by default on Android. Starting with Kotlin 2.1.20, this feature will be enabled by default on all platforms.

Remember to check if you set this option using `freeCompilerArgs`. This method can cause the build to fail when used alongside the plugin, due to an option being effectively set twice.

Breaking changes and deprecations

- To align Kotlin Multiplatform with upcoming changes in Gradle, we are phasing out the `withJava()` function. Java source sets are now created by default ↗. If you use the Java test fixtures ↗ Gradle plugin, upgrade directly to Kotlin 2.1.21 to avoid compatibility issues.
- The JetBrains team is proceeding with the deprecation of the `kotlin-android-extensions` plugin. If you try to use it in your project, you'll now get a configuration error, and no plugin code will be executed.
- The legacy `kotlin.incremental.classpath.snapshot.enabled` property has been removed from the Kotlin Gradle plugin. The property used to provide an opportunity to fall back to a built-in ABI snapshot on the JVM. The plugin now uses other methods to detect and avoid unnecessary recompilations, making the property obsolete.

Documentation updates

The Kotlin documentation has received some notable changes:

Revamped and new pages

- Kotlin roadmap – see the updated list of Kotlin's priorities on language and ecosystem evolution.
- Gradle best practices page – learn essential best practices for optimizing your Gradle builds and improving performance.
- Compose Multiplatform and Jetpack Compose ↗ – an overview of the relation between the two UI frameworks.
- Kotlin Multiplatform and Flutter ↗ – see the comparison of two popular cross-platform frameworks.
- Interoperability with C – explore the details of Kotlin's interoperability with C.
- Numbers – learn about different Kotlin types for representing numbers.

New and updated tutorials

- Publish your library to Maven Central ↗ – learn how to publish KMP library artifacts to the most popular Maven repository.
- Kotlin/Native as a dynamic library – create a dynamic Kotlin library.
- Kotlin/Native as an Apple framework – create your own framework and use Kotlin/Native code from Swift/Objective-C applications on macOS and iOS.

How to update to Kotlin 2.1.20

Starting from IntelliJ IDEA 2023.3 and Android Studio Iguana (2023.2.1) Canary 15, the Kotlin plugin is distributed as a bundled plugin included in your IDE. This means that you can't install the plugin from JetBrains Marketplace anymore.

To update to the new Kotlin version, change the Kotlin version to 2.1.20 in your build scripts.